

- HANDS-ON WORKSHOP • 2 HOURS
- LEVEL 02

Advanced FinOps Plugin.

Extending a Markdown plugin with MCP integrations — live market data, client documents from disk, and persistent memory across sessions.



— PLUGIN STACK

mcp.

- tavily • live market data
- memory • persistent graph
- native • file tools (built-in)

By the end of the session.

01

A fully wired plugin

finops-advanced with two MCP servers and Claude Code's native file tools threaded through every command.

02

Three new commands

Portfolio review, live market briefings, and compliant retirement projections — each grounded in real data, not the model's memory.

03

Agents that remember

Compliance checks verified against ASIC in real time. Client context carried from one session to the next.

• SETUP • 15 MIN

• WALKTHROUGH • 25 MIN

• EXERCISES • 60 MIN

• REPORT + WRAP • 20 MIN

• WHAT'S DIFFERENT

Level 1 → Level 2.

FEATURE	FINOPS-BASIC	FINOPS-ADVANCED
MCP servers	None	tavily · memory
Market data	Static, from Claude's training	Live, via tavily with citations
Client documents	Manual paste only	Read from disk via native tools
Client history	Starts fresh each session	Persists via memory MCP
Compliance check	Knowledge-based	Verified against ASIC.gov.au
New commands	—	portfolio-review · market-briefing · retirement-projection
New agents	—	portfolio-analyst · market-researcher

• BEFORE WE START

Three things on your machine.

CLI

Claude Code

`claude --version` should print a version. If not, install it before we go further.

API KEY

Tavily, free tier

Sign up at **app.tavily.com** — takes about two minutes. We'll set `TAVILY_API_KEY` in a moment.

DUMMY DATA

Sample client docs

A few PDFs, CSVs or text files. Real or fake — we'll generate a demo client folder in Exercise 1 either way.

— HEADS UP

- ✓ No code is required — this is configuration only. The plugin is plain Markdown.
- ✓ Everything runs locally; memory lives in your home directory, not the cloud.

01

• PART ONE • 15 MINUTES

Setup.

Two steps. Register the MCP servers globally once, then launch Claude Code with the plugin directory on every session.

Register the MCP servers.

`--plugin-dir` loads commands and agents but does **not** read `.mcp.json` from the plugin folder. MCP servers must be registered globally.

Set your API key

```
# macOS / Linux
$ export TAVILY_API_KEY=your_key_here

# persist it
$ echo 'export TAVILY_API_KEY=...' >> ~/.zshrc
```

Register globally

```
$ claude mcp add tavily \
  --scope user \
  --env TAVILY_API_KEY=$TAVILY_API_KEY \
  -- npx -y tavily-mcp

$ claude mcp add memory \
  --scope user \
  -- npx -y @modelcontextprotocol/server-memory
```

— TIP

Both servers run via `npx` — no global install, no Docker. Node must be on your PATH.

Launch with the plugin.

MACOS / LINUX

```
$ claude \  
  --plugin-dir /path/to/level2/  
plugin-level2/finops-advanced
```

WINDOWS • POWERSHELL

```
> claude `  
  --plugin-dir C:\path\to\level2\  
plugin-level2\finops-advanced
```

Example absolute path

```
$ claude --plugin-dir /media/daghan/DATA/Daghan/git/EDA-repository/agent-experiments/claudeplugin/level2/plugin-level2/finops-advanced
```

— WHY THIS EVERY TIME

- ✓ Plugin commands and agents are loaded per session — the global registry only holds MCP servers.
- ✓ Alias it: `alias finops="claude --plugin-dir /path/to/finops-advanced"`

Verify the servers loaded.

Inside the Claude Code session, run:

```
> /mcp
```

You should see both servers in the list. If either is missing, return to step 1 and check the registration scope.

Expected output

- **tavily** – connected
- **memory** – connected

TROUBLESHOOTING

- `claude mcp list` · see what's registered
- `claude mcp remove [name]` · remove & re-add

02

• PART TWO • 10 MINUTES

Plugin anatomy.

A tour of the folder. What's new in v2, what's been enhanced, and what we left alone.

● THE FOLDER

Plugin structure.

```

plugin-level2/finops-advanced/
├── .claude-plugin/plugin.json      plugin metadata (v2.0)
├── .mcp.json                      MCP server config
├──
├── skills/
│   ├── compliance-language/      + tavily for ASIC lookups
│   ├── client-meeting-prep/      + native file + memory
│   ├── email-tone/              unchanged
│   └── market-awareness/        NEW · live context
├──
├── commands/
│   ├── draft-report.md           + reads portfolio
│   ├── prep-meeting.md          + memory recall
│   ├── weekly-summary.md        + memory x-ref
│   ├── portfolio-review.md      NEW
│   ├── market-briefing.md      NEW
│   ├── retirement-projection.md NEW
│   ├── generate-report.md       NEW · .docx
│   └── setup-test-client.md     NEW · demo data
├──
├── agents/
│   ├── compliance-reviewer.md   + live ASIC
│   ├── meeting-summariser.md   + memory write
│   ├── portfolio-analyst.md     NEW
│   ├── market-researcher.md    NEW
│   └── report-generator.md      NEW
├──
└── scripts/
    ├── fill_report.py           NEW · .docx filler
    └── sample_data.json         NEW · test payload

```

- **NEW** — added in v2
- **Enhanced** — wired to MCP
- **Unchanged** — same as v1

THE SHAPE OF V2

5 new commands, 3 new agents, 1 new skill — all wired to the same two MCP servers and Claude Code's native file tools.

03

• PART THREE • 15 MINUTES

Three integrations.

Files on disk • live web • persistent memory. What each one does, who uses it, and what to run to see it work.

Client documents from disk.

Claude Code reads files directly — no MCP needed. The Read tool is built in.
We just point commands at absolute paths.

Used by

- **/portfolio-review** — reads statement files
- **/prep-meeting** — fact-finds + SOA drafts
- **/draft-report** — portfolio data when path provided

— TRY IT NOW

Spin up the demo client folder so the next exercise has something to read.

Generate a test client

```
> /finops-advanced:setup-test-client  
> /clear
```

`/clear` wipes context so the portfolio gets read fresh next call.

THEN RUN

```
> /finops-advanced:portfolio-review \  
  ~/Documents/clients/smith-jane/portfolio.txt \  
  balanced
```

Live market data.

Gives Claude the ability to search the web for current market information — with source URLs and dates next to every figure.

Used by

- **/market-briefing** — ASX, RBA, CPI, sectors
- **compliance-reviewer** — verifies ASIC rules
- **market-researcher** — pulls briefing data
- **market-awareness** skill — surfaces context

— OBSERVE

- ✓ Every figure carries a source URL and a date.
- ✓ Talking points are grounded in retrieved data, not inferred.
- ✓ Data gaps are noted — never estimated.

Try it

```
> /finops-advanced:market-briefing \  
healthcare and technology sectors
```

2

MCP SERVERS IN V2

4

PLACES TAVILY PLUGS IN

Client continuity.

A persistent knowledge graph that survives across sessions. Action items, risk profiles, and meeting context outlive a single chat.

Used by

- **meeting-summariser** — writes actions after each meeting
- **/prep-meeting Jane Smith** — recalls prior actions
- **/weekly-summary** — surfaces recurring themes

The shape of a memory write

```
> /finops-advanced:summarise-meeting \  
  Jane Smith — meeting 3 Mar. \  
  Discussed CBA concentration risk. \  
  Action: adviser to prepare rebalancing \  
  options by 10 Mar.
```

...THEN, IN A BRAND-NEW CLAUDE SESSION

```
> /finops-advanced:prep-meeting Jane Smith  
> /finops-advanced:weekly-summary Jane Smith
```

— WHAT YOU'LL OBSERVE

The CBA concentration risk and 10 Mar action item are recalled — even though you closed the prior session.

04

• PART FOUR • 60 MINUTES • HANDS-ON

Five exercises.

Portfolio review • live market briefing • retirement
projection • memory continuity across sessions • real-time
compliance verification.

01

EXERCISE • 12 MIN

Portfolio review.

Run

```
> /finops-advanced:portfolio-review \  
  ~/Documents/clients/smith-jane/\  
portfolio.txt \  
  balanced
```

What's happening under the hood

- Claude reads the file with its native Read tool — no MCP call
- **portfolio-analyst** agent computes allocation vs target
- tavily is queried for current RBA rate & ASX 200

— OBSERVE

- ✓ Allocation-vs-target table is generated cleanly
- ✓ Concentration flags appear for any holding over 10%
- ✓ Current RBA rate and ASX 200 sit in the market-context block
- ✓ Recommendations cite both the file and live data

— STRETCH

Re-run with **growth** as the risk profile and watch the rebalancing logic change.

02

EXERCISE • 10 MIN

Live market briefing.

Run — broad

```
> /finops-advanced:market-briefing
```

...then with a sector focus

```
> /finops-advanced:market-briefing \  
  REITs and infrastructure
```

UNDER THE HOOD

market-researcher agent fans out tavily queries — ASX 200, RBA, CPI, sector indices — then composes a briefing pack with inline citations.

— OBSERVE

- ✓ Every data point includes a source URL and date
- ✓ Adviser talking points are grounded in the retrieved data, not inferred
- ✓ Data gaps are explicitly noted, never estimated
- ✓ The narrative reads like a real morning briefing, not a search dump

— STRETCH

Ask a follow-up: *“Which sector has the most negative momentum this week?”* — note the agent re-runs tavily rather than guessing.

03

EXERCISE • 12 MIN

Retirement projection.

Run

```
> /finops-advanced:retirement-projection \  
48 65 320000 18000 growth
```

ARGUMENTS

48	current age
65	retirement age
320000	current balance (AUD)
18000	annual contribution
growth	risk profile

— OBSERVE

- ✓ Three scenarios — low, mid, high — with distinct return assumptions
- ✓ All projections in today's dollars (real terms)
- ✓ Required disclaimers appear inline, not as an afterthought
- ✓ Contribution-sensitivity table shows the impact of saving more

— WHY THIS MATTERS

Notice how *compliance language* is woven into the projection — the agent doesn't bolt it on at the end. This is the *compliance-language* skill firing in the background.

04

EXERCISE • 14 MIN • ACROSS SESSIONS

Memory continuity.

01

Capture a meeting note

Run `/summarise-meeting` with notes that include a client name (e.g. **Jane Smith**) and explicit action items.

02

Close Claude Code completely

Quit the process. Open a fresh terminal. The memory MCP holds the graph on disk — it survives.

03

Relaunch with the plugin

Same `claude --plugin-dir ...` command as before.

04

Prep for Jane Smith's next meeting

```
> /finops-advanced:prep-meeting Jane Smith
```

— OBSERVE

- ✓ Prior action items and client context are recalled — even from the previous session
- ✓ The carried-over items show up explicitly under "open follow-ups"

05

EXERCISE • 12 MIN • REQUIRES TAVILY KEY

Compliance verification.

Run

```
> /review-draft [paste a draft client  
email with some  
borderline language]
```

TRY LANGUAGE LIKE

- *"guaranteed returns"*
- *"this is the safest investment"*
- *"low risk, high reward"*

— OBSERVE

- ✓ compliance-reviewer first searches ASIC.gov.au to verify current requirements
- ✓ The report cites the search date and notes any regulatory updates
- ✓ Issues are flagged with suggested replacement language

— WHY THIS MATTERS

The agent doesn't rely on its training-data view of ASIC rules. It checks the live source — every time.

05

• PART FIVE • 15 MINUTES

Generate report.

From three data sources — memory, disk, and live market — to a filled Word document a client can open in Office.

• GENERATE-REPORT PIPELINE

How it works.

01

Template

You supply a `.docx` file with `{{FIELD}}` placeholders.

02

Gather data

memory for risk profile · **disk** for portfolio · **tavily** for RBA / ASX / CPI.

03

Build payload

report-generator agent
assembles a JSON object keyed
by placeholder name.

04

Fill & save

`fill_report.py` replaces every
placeholder, names the output by client +
period, writes the `.docx`.

— BEFORE YOU RUN IT

Copy `NGC-2025-0041_Q4-2025_quarterly-review.docx` into `~/Documents/clients/templates/`.

• TEMPLATE FORMAT

Placeholders.

Use `{{UPPERCASE_WITH_UNDERSCORES}}` in your Word document. The script replaces every match.

PLACEHOLDER	EXAMPLE VALUE
<code>{{CLIENT_NAME}}</code>	James and Helen Nguyen
<code>{{CLIENT_RISK_PROFILE}}</code>	Balanced
<code>{{PORTFOLIO_TOTAL_VALUE}}</code>	\$842,300
<code>{{RBA_CASH_RATE}}</code>	4.10%
<code>{{ASX200_LEVEL}}</code>	8,142
<code>{{REPORT_DATE}}</code>	4 March 2026
<code>{{ADVISER_NAME}}</code>	Sarah Chen
<code>{{DISCLAIMER}}</code>	standard disclaimer text

OUTPUT NAMING

`[CLIENT_REF]_[PERIOD]_[TYPE].docx`

e.g. NGC-2025-0041_Q4-2025_quarterly-review.docx

— FULL LIST

✓ `scripts/sample_data.json` has the complete field reference.

• TWO MODES

Running the report.

QUICK MODE

Pre-built JSON

Skips data collection. Useful to verify a new template before wiring up live data.

```
> /finops-advanced:generate-report \  
  ~/Documents/clients/templates/  
NGC-2025-0041_Q4-2025_quarterly-review.docx \  
  --data scripts/sample_data.json
```

FULL MODE

Agent gathers everything

Pulls memory · disk · tavily into one pass. The slow path, but the production path.

```
> /finops-advanced:generate-report \  
  ~/Documents/clients/templates/  
NGC-2025-0041_Q4-2025_quarterly-review.docx \  
  Jane Smith
```

— WHAT TO LOOK FOR

- ✓ The agent reports the output file path on completion.
- ✓ Any unfilled `{{PLACEHOLDERS}}` are listed so you can supply missing data — nothing silently goes blank.

• WHEN THINGS GO SIDEWAYS

Troubleshooting.

”MCP server not found”

```
$ claude mcp list
$ claude mcp remove [name]
```

Check what's registered. Remove and re-add if the entry is stale.

Tavily returns nothing

```
$ echo $TAVILY_API_KEY
```

Confirm the key is exported in this shell. Check quota at app.tavily.com.

Can't find a client file

Use **absolute paths** (`~/Documents/...`). Claude Code's Read tool needs read permissions for your user — no MCP involved.

Memory doesn't persist

Confirm memory is in `claude mcp list` . The graph lives in the MCP server's data dir — if it's been wiped or reinstalled, you'll start fresh.

• RECAP

What you built.

01

A Markdown plugin
extended with MCP

No custom code. Just
configuration.

02

Live market data via
Tavily

Citations & dates on
every figure.

03

Client docs from
disk

Native Read tool — no
extra MCP.

04

Cross-session
memory

Action items survive restart.

05

Compliant,
grounded
commands

Portfolio · retirement
· briefings.

5

NEW COMMANDS

3

NEW AGENTS

2

MCP SERVERS

0

LINES OF CUSTOM CODE

Only add an MCP server when Claude doesn't already have the capability.

File access, bash execution, and code editing are built into Claude Code. MCP is for what Claude *can't* do natively — live web search, persistent cross-session memory, third-party services.



ship.

You've extended a Markdown plugin into a live-data, memory-aware adviser workflow.

Zero lines of custom code.